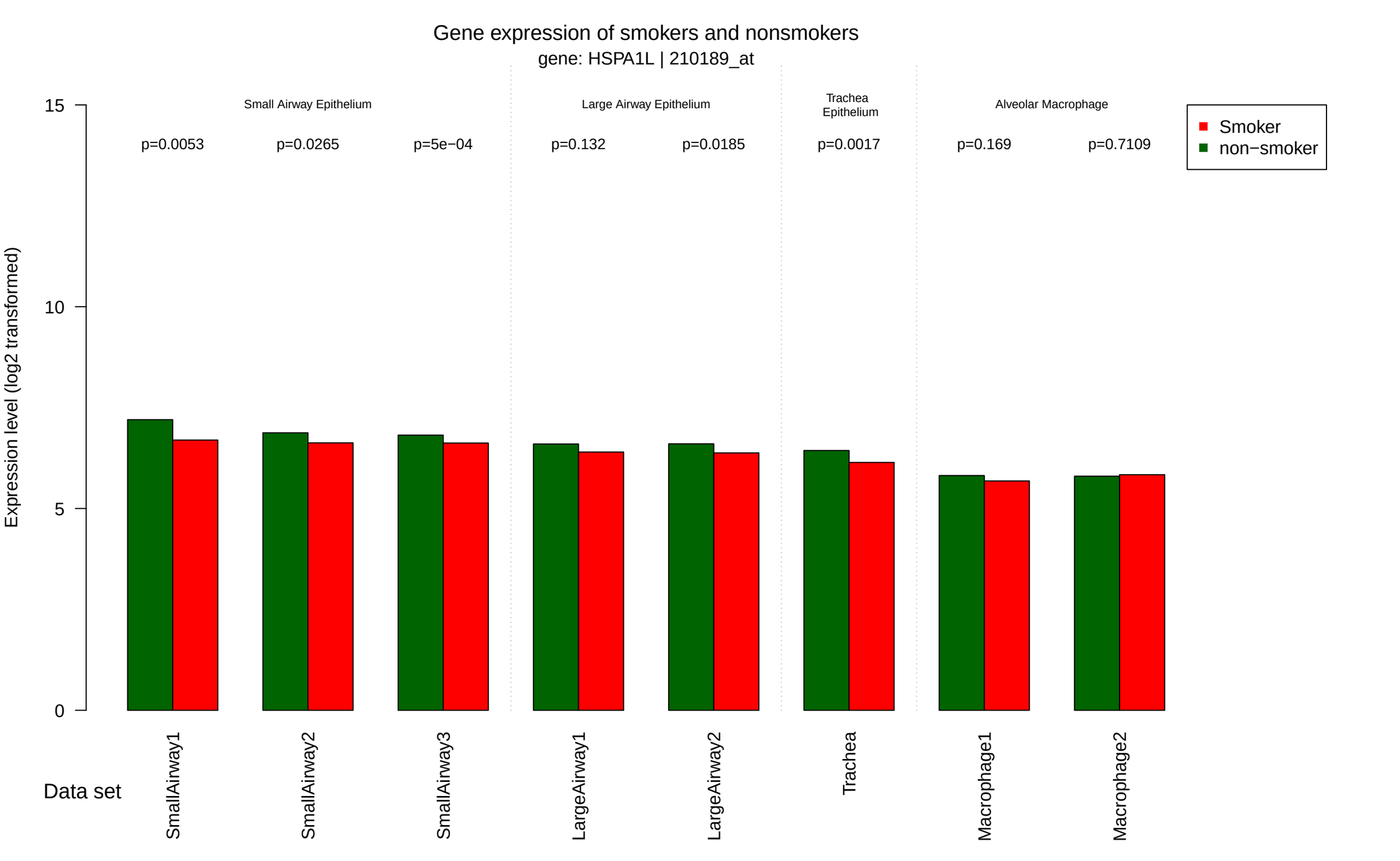
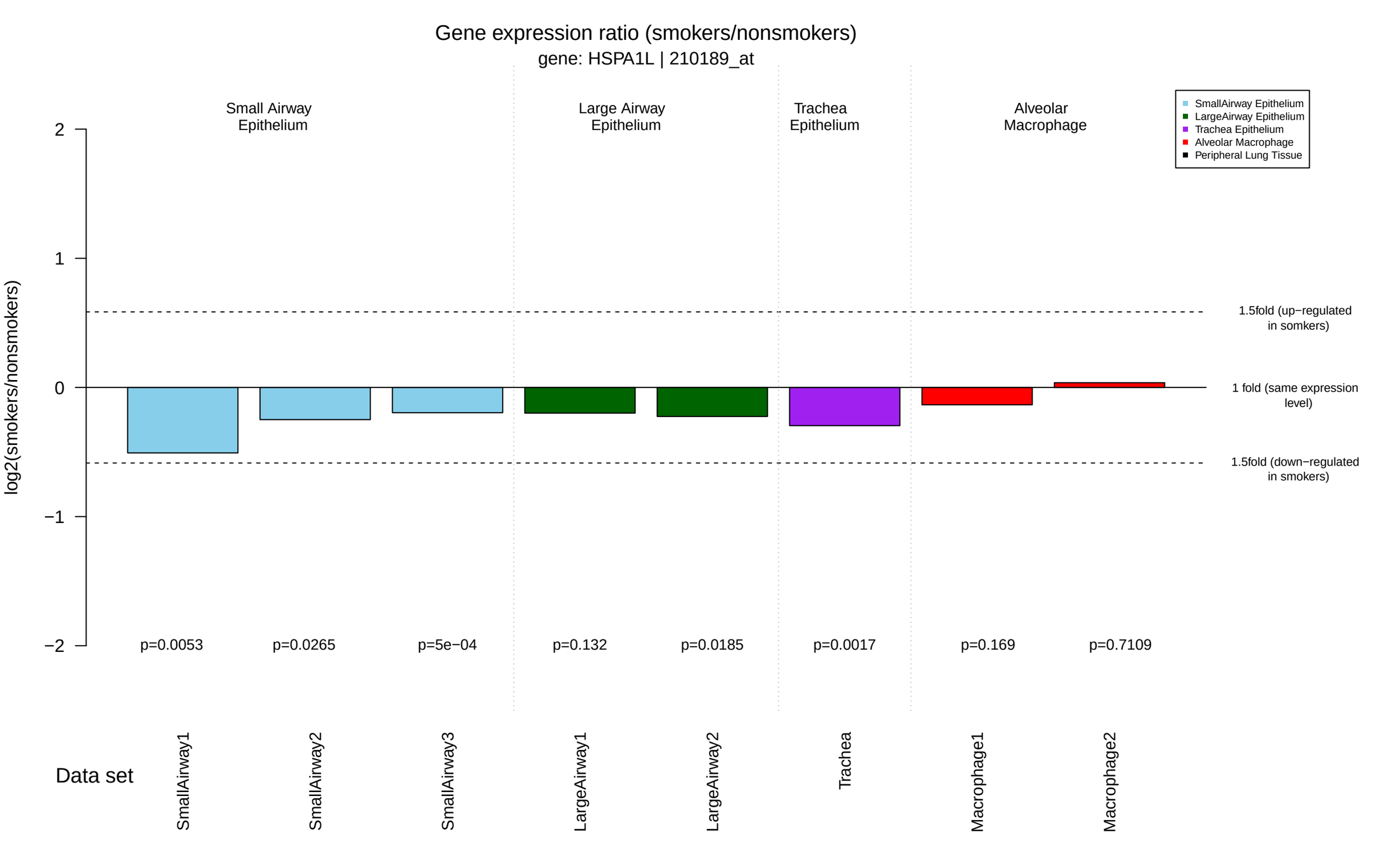
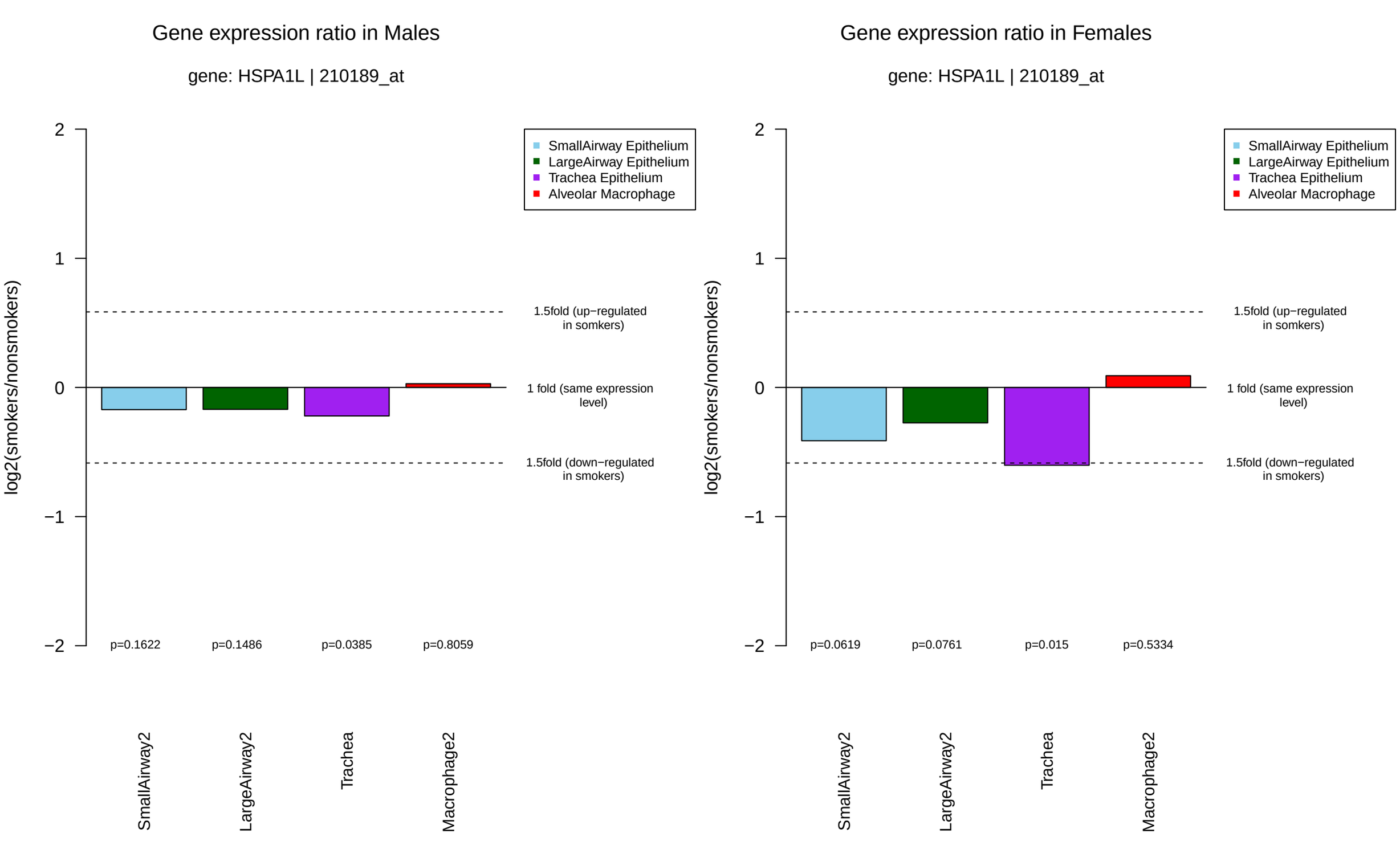
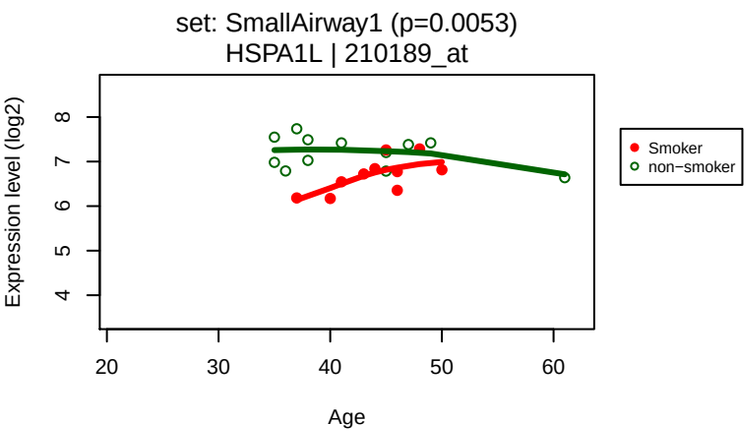
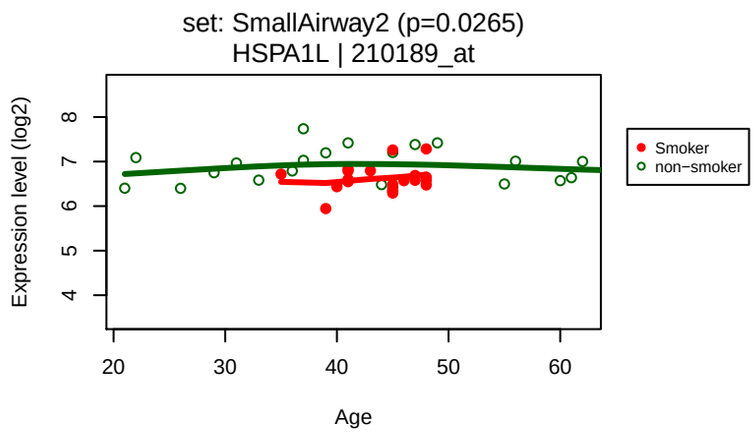
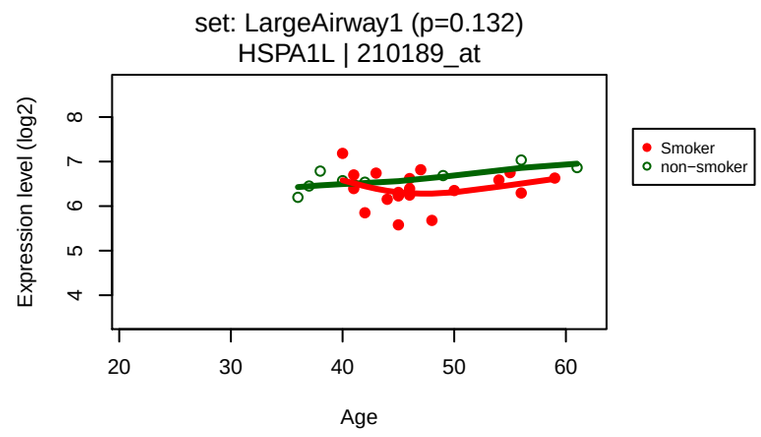

Gene expression of smokers and nonsmokers
gene: HSPA1L | 210189_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0053
p=0.0265
p=5e−04
p=0.132
p=0.0185
p=0.0017
p=0.169
p=0.7109
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

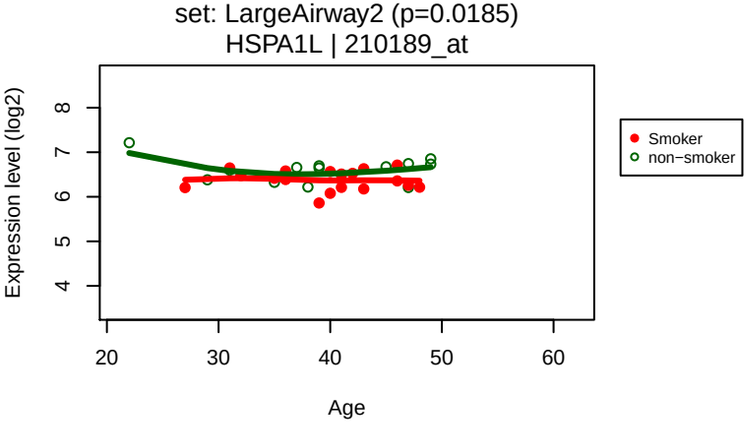

Gene expression ratio (smokers/nonsmokers)
gene: HSPA1L | 210189_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0053
p=0.0265
p=5e−04
p=0.132
p=0.0185
p=0.0017
p=0.169
p=0.7109
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

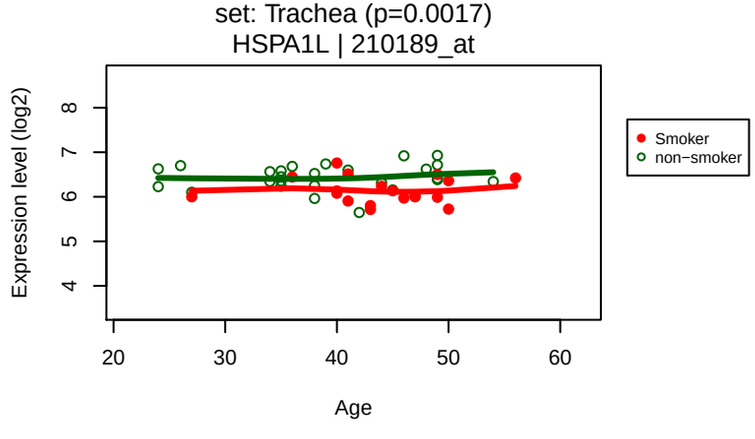

Gene expression ratio in Males
gene: HSPA1L | 210189_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1622
p=0.1486
p=0.0385
p=0.8059
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA1L | 210189_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0619
p=0.0761
p=0.015
p=0.5334
SmallAirway2
LargeAirway2
Trachea
Macrophage2

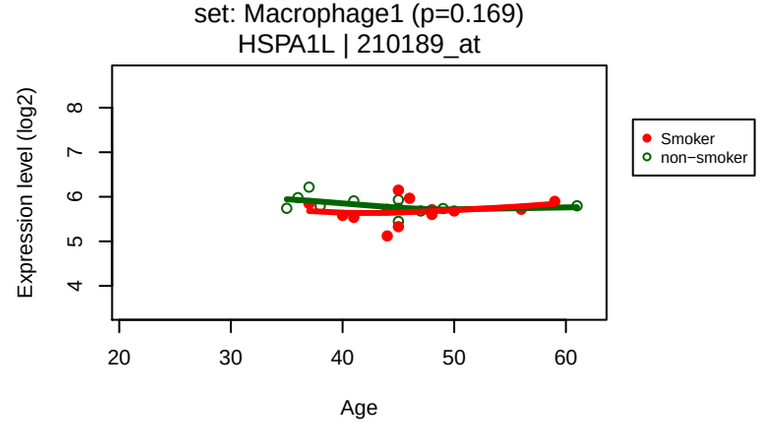

set: SmallAirway1 (p=0.0053)
HSPA1L | 210189_at
Expression level (log2)
Age
Smoker
non−smoker

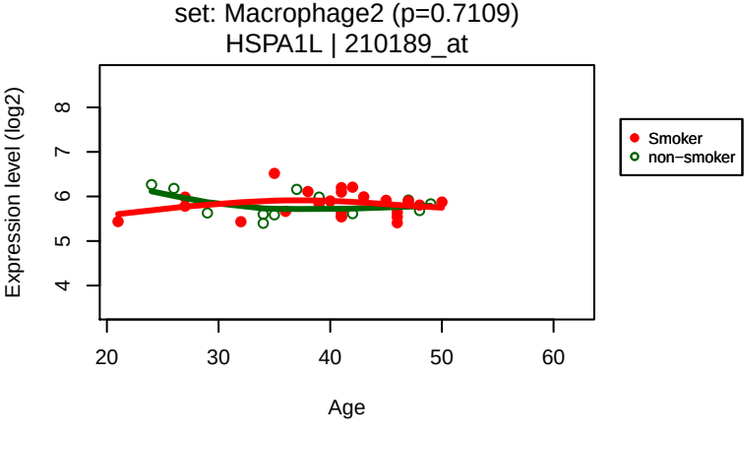

set: SmallAirway2 (p=0.0265)
HSPA1L | 210189_at
Expression level (log2)
Age
Smoker
non−smoker

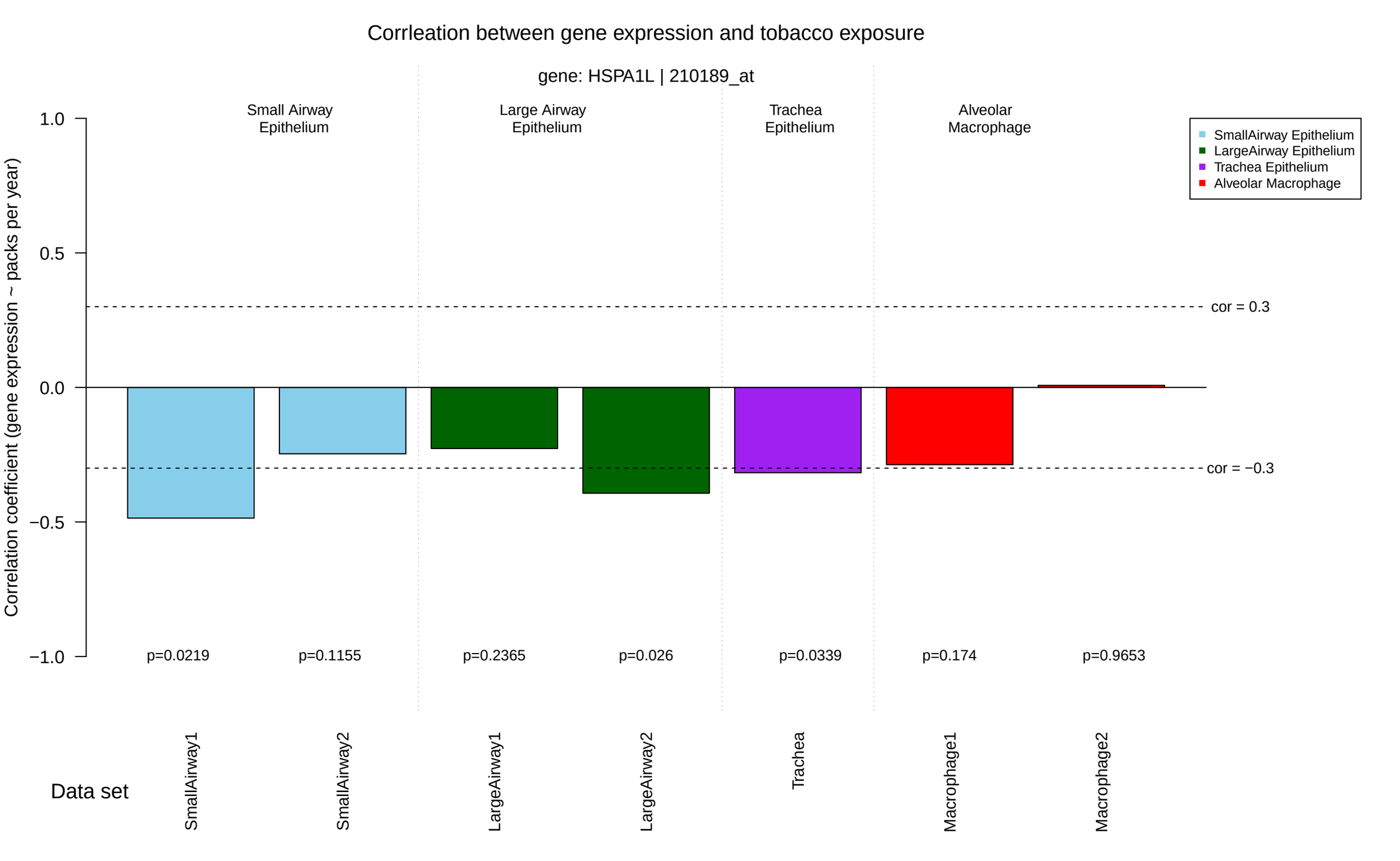

set: LargeAirway1 (p=0.132)
HSPA1L | 210189_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0185)
HSPA1L | 210189_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0017)
HSPA1L | 210189_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.169)
HSPA1L | 210189_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7109)
HSPA1L | 210189_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA1L | 210189_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0219
p=0.1155
p=0.2365
p=0.026
p=0.0339
p=0.174
p=0.9653
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2